

Product Requirement Document (PRD)

Project Name: VerifyMed

System Subtitle: Real-Time Multi-Agent Drug Authentication Infrastructure

Document Version: 1.1.0

Date: May 21, 2026

Status: Approved / Ready for Implementation

Target Audience: Engineering Leads, Product Managers, Educational Workshop Facilitators, AI Architects

Document Control & Metadata

Version	Date	Author	Description of Changes	Status
1.0.0	May 21, 2026	Product Team	Initial draft for stakeholder review.	Draft
1.1.0	May 21, 2026	Technical Architecture	Integrated Groq-specific inference pipelines, model-to-task routing matrices, and exhaustive step-by-step edge-case failovers.	Approved

1. Executive Summary & Core Objective

1.1 Problem Statement

In Sub-Saharan Africa, counterfeit pharmaceuticals represent a critical and deadly humanitarian crisis. The World Health Organization (WHO) estimates that **over 100,000 lives**

are lost annually in Africa due to counterfeit, sub-therapeutic, or contaminated drugs. Nigeria is one of the most severely affected markets.

Consumers and healthcare providers purchase medications from pharmacies and open-air markets with no quick, deterministic method to verify if a drug packaging represents a legitimate batch, a stolen shipment, or a chemically inert counterfeit (such as chalk or starch packaged as an antimalarial).

1.2 The Vision

VerifyMed is a high-speed, multi-agent AI framework designed to run immediate forensic verifications on medicine. By taking a phone camera photograph of a drug box or typing in raw batch variables, a user initiates a multi-layered validation sequence. Four specialized, parallel AI agents run deep verification loops—spanning regulatory registration, global safety intelligence, and physiological plausibility—synthesizing their findings into a single, highly actionable, probabilistic verdict in **under 5 seconds**.

1.3 Strategic Purpose as an Educational Blueprint

While highly viable as an enterprise-grade utility, VerifyMed is engineered specifically to act as the ultimate **teaching catalyst** for AI engineers and computer science students. It demonstrates the following major paradigms in modern AI system design:

- **Asynchronous Parallel Agentic Execution:** Moving away from slow, sequential chain-of-thought steps toward concurrent agent execution to slash user latency.
- **Model-to-Task Tuning (Cost-Speed Right-Sizing):** Teaching developers how to map tasks to cheap micro-models (8B parameter sizes) for high-speed deterministic operations, while reserving flagship reasoning engines (120B+) strictly for the final logical synthesis.
- **Vision-to-Structured Data Transformation (Multimodal OCR):** Implementing rigid schema extraction using visual token structures.
- **RAG (Retrieval-Augmented Generation) & Live Tool Use:** Integrating external APIs, news feeds, and scientific literature search directly into independent agent layers.

2. User Personas & High-Level Workflows

VerifyMed caters to three primary target personas:

USER PERSONAS		
1. The Local Pharmacist	2. The Patient/Consumer	3. The Workshop Developer
Verifies incoming bulk inventory in urban/rural clinical centers.	Snaps a photo of a newly purchased antimalarial box at point of purchase.	Builds, debugs, and views the internal agent state transitions and latency.

2.1 The Clinical Pharmacist (Primary User)

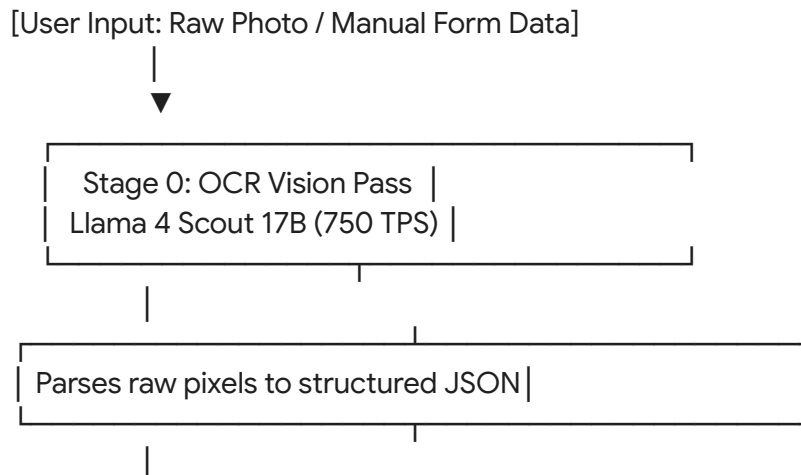
- **Persona Profile:** Working in high-throughput environments in Lagos, Abuja, or Kano. Needs to process dozens of boxes rapidly. No tolerance for slow load times, high error rates, or complex interfaces.
- **Workflow:**
 1. Launches the VerifyMed responsive web app on a mobile device.
 2. Holds a drug box in front of the camera and triggers the **Image Capture** mode.
 3. Watches the parallel agent diagnostic terminal resolve discrepancies.
 4. Saves the record card with a digital timestamp and validation code.

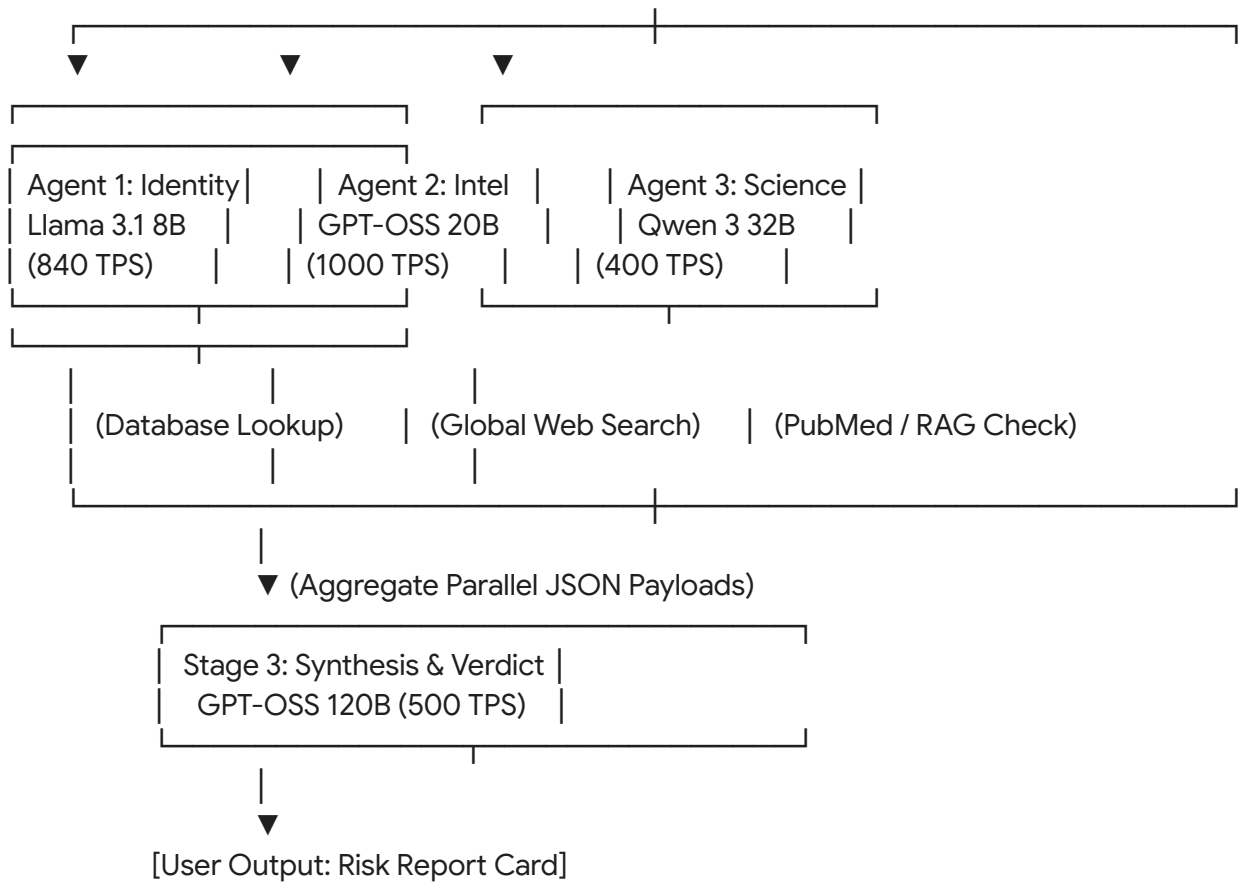
2.2 The Student / Workshop Developer (Technical User)

- **Persona Profile:** Engineering student or experienced developer attending a hackathon or Masterclass. Needs to understand agent state-passing, latency optimization, and JSON serialization.
- **Workflow:**
 1. Uses VerifyMed's visual dashboard to run preset simulations.
 2. Explores the detailed telemetry stream of each independent model.
 3. Toggles between manual JSON/Text input and multimodal image uploads to test parsing performance.
 4. Inspects the underlying Product Requirement Document (PRD) to cross-reference architectural expectations.

3. System Architecture & Model-to-Task Routing

To achieve near-instantaneous processing (target sub-5 second response times), VerifyMed does not run all actions on a single monolithic frontier model. Instead, it utilizes a highly optimized **Model-to-Task Routing Grid** running on Groq hardware.





3.1 Model-to-Task Specifications

3.1.1 Stage 0: Multimodal OCR Gateway

- **Assigned Model:** meta-llama/llama-4-scout-17b-16e-instruct
- **Key Capability:** Visual reasoning, low-latency spatial character mapping.
- **Functional Description:** Processes the raw camera byte stream. Locates, extracts, and serializes the physical text found on the medicine box.
- **Target Output Schema:**

```

{
  "brand_name": "string",
  "batch_number": "string",
  "manufacturer": "string",
  "active_ingredients": ["string"],
  "stated_therapeutic_use": "string"
}

```

3.1.2 Agent 1: The Identity Agent

- **Assigned Model:** llama-3.1-8b-instant

- **Key Capability:** Lightning-fast deterministic tool calling.
- **Functional Description:** Ingests the output of Stage 0. It queries the NAFDAC database mock endpoints and the WHO Prequalification directory to check if the manufacturer registration and the batch sequence align with official records.

3.1.3 Agent 2: The Intelligence Agent

- **Assigned Model:** openai/gpt-oss-20b (or equivalent open-weights search model)
- **Key Capability:** Integrated internet search and temporal contextual extraction.
- **Functional Description:** Executes live Google Search, GDELT, or news wire queries. Searches for terms like: "[Manufacturer] + recall", "[Brand Name] + counterfeit + Nigeria". Extracts raw warning files from global custom agencies.

3.1.4 Agent 3: The Science Agent

- **Assigned Model:** qwen/qwen3-32b
- **Key Capability:** Deep academic, medical, and chemistry-focused logical reasoning.
- **Functional Description:** Compares the stated active ingredients against the intended therapeutic uses declared on the packaging. Analyzes physiological compatibility. Flag cases where a drug claims to treat a condition that is biologically inconsistent with its physical chemical properties (e.g., an antimalarial claiming to cure a bacterial infection or viral flu).

3.1.5 Agent 4: The Verdict Agent

- **Assigned Model:** openai/gpt-oss-120b
- **Key Capability:** High-level logical synthesis, nuance arbitration, and ethical constraint compliance.
- **Functional Description:** Consumes the parallel outputs of Agents 1, 2, and 3. Resolves conflicting evidence, evaluates systemic risks, calculates a final probabilistic confidence score, and structures the final patient-safe notification card.

4. Functional Requirements (FR)

FR-1: Multimodal Ingestion (The Front Gate)

- **Description:** The system must allow users to upload an image of a medicine container/box or manually type structured inputs.
- **Input Types:** PNG, JPG, WebP (up to 10MB) or form field input.
- **Vision Fallback:** If the vision model fails to parse structural parameters due to image blur, the system must immediately present a manual entry interface to prevent user obstruction.

FR-2: Real-Time Parallel Execution Engine

- **Description:** To guarantee speed, Agent 1, Agent 2, and Agent 3 must run asynchronously in parallel, non-blocking threads.
- **State Aggregation:** The system must implement a barrier synchronization pattern, where

Agent 4 (Verdict) is only triggered once all three concurrent agent threads complete their output executions.

- **Execution Telemetry:** The system must capture and expose latency metrics (tokens per second, total time taken per agent) for debugging and pedagogical feedback.

FR-3: Streaming Thought Terminal

- **Description:** The frontend interface must stream the intermediate reasoning tokens ("internal monologues") of Agents 1, 2, and 3 simultaneously in clear, distinct, side-by-side terminal logs.
- **Rationale:** Standard loading spinners trigger higher user drop-offs. Streaming live agent thoughts dramatically increases perceived performance and serves as a vital learning portal for developers.

FR-4: Probabilistic Risk Assessment Summary

- **Description:** The output must present a beautifully color-coded safety report card categorizing the compound's risk profile:
 -  **CRITICAL FAILURE / FAKE (Score: 0 – 20%)**: Clear presence of fraudulent manufacturer data, active safety recalls, or dangerous medical fallacies.
 -  **SUSPICIOUS / HIGH RISK (Score: 21 – 79%)**: Mismatched registration details, spelling anomalies on packaging, or active warnings in nearby jurisdictions.
 -  **VERIFIED RECORD (Score: 80 – 100%)**: Perfect alignment with registry details, zero reported news warnings, and impeccable chemical logic.
- **Consensus Logs:** The UI must display the explicit evidence points extracted by each agent to support the Verdict Agent's synthesized score.

5. Non-Functional Requirements (NFR)

NFR-1: End-to-End Latency

- **Specification:** Under normal network conditions, the cumulative duration from user input submission to final verdict presentation must not exceed **5.0 seconds**.
- **Optimization:** Achieved through parallel asynchronous execution of middle-tier agents on ultra-low-latency Groq LPUs.

NFR-2: Strict Data Validation & Serialization

- **Specification:** All inter-agent data passing must utilize rigid JSON schemas. Any agent output that fails validation must trigger a graceful schema repair auto-prompt, ensuring that Agent 4 always receives structured payloads.

NFR-3: Operational Safety & Moral Liability

- **Specification:** To prevent the system from being held liable in clinical emergencies, the

Verdict Agent must **never** output absolute, binary assertions (e.g., *"This medicine is 100% genuine and safe"*). All statements must use precise, probabilistic data-centric terms (e.g., *"This record matches official registration profiles"*).

- **Visual Warning Guardrail:** The mobile dashboard must permanently display a prominent, highly readable medical safety disclaimer.

NFR-4: Universal Responsive Design

- **Specification:** Since pharmacists and consumers in Nigeria primarily use mid-to-low tier mobile smartphones, the interface must be highly optimized for mobile devices, utilize relative widths (avoiding fixed horizontal spacing), and adapt to extreme lighting environments.

6. Detailed Agent Prompt & Output Specifications

To ensure perfect educational implementation, this section outlines the exact system prompts and response schemas required for each active agent node.

6.1 Agent 1: The Identity Agent (Llama-3.1-8B-Instant)

[System Prompt]

You are Agent 1 (Identity Agent) for VerifyMed. Your task is to query regulatory drug registries (NAFDAC, WHO Prequalification) to check if the brand name, manufacturer, and batch sequence represent registered medical records.

Input Payload: {brand_name, manufacturer, batch_number}

Output Form: Write a clear, concise log outlining match status. Be highly analytical.

- **Response Format (JSON output):**

```
{
  "manufacturer_registered": true,
  "batch_format_valid": true,
  "registry_match_found": true,
  "flagged_anomalies": ["None"]
}
```

6.2 Agent 2: The Intelligence Agent (GPT-OSS-20B-Search)

[System Prompt]

You are Agent 2 (Intelligence Agent) for VerifyMed. You execute external queries to detect public warnings, drug recalls, police port raids, or Interpol warnings related to the drug brand name or batch.

Input Payload: {brand_name, batch_number, manufacturer}

Output Form: Detail matching global recall notices, region-specific counterfeit alerts, or state reports.

6.3 Agent 3: The Science Agent (Qwen-3-32B)

[System Prompt]
You are Agent 3 (Science Agent) for VerifyMed. You are an expert pharmacologist. Examine the stated active ingredients and cross-reference them with the intended therapeutic treatments written on the package.
Assess logical efficacy. Highlight any pharmacological inconsistencies (e.g., paracetamol claiming to cure a bacterial infection, or antimalarials claiming to cure viral respiratory flu).
Input Payload: {active_ingredients, intended_treats}

6.4 Agent 4: The Verdict Agent (GPT-OSS-120B)

[System Prompt]
You are Agent 4 (Verdict Agent), the supreme orchestrator of VerifyMed. You will receive the individual logs and structured outputs from Agent 1 (Identity), Agent 2 (Intelligence), and Agent 3 (Science).
Your task is to:

1. Parse the three reports for any conflicting arguments.
2. Formulate a final risk assessment.
3. Calculate a safety confidence score from 0% to 100%.
4. Draft a highly clear, supportive, and protective summary for the user.
5. Adhere strictly to the safety disclaimer rules: Use data-probabilistic language. Never state "safe to drink/consume" or "100% genuine".

7. Edge Cases & System Failures (Error Matrix)

Failure Scenario	Technical Detection Method	System Auto-Mitigation Strategy
Stage 0 OCR Failure (Blurry photo, cut-off label)	Vision API returns an empty schema or JSON extraction fails with confidence rating < .	Instantly toggle the interface to manual form-field mode with a warning: "Camera could not parse text. Please check the text input below to proceed."
Agent API Rate-Limitation / Timeout	LPU inference thread fails to	Run an automated retry with exponential backoff (

	resolve in > seconds.	0.5s → 1s →). If a sub-agent (e.g., Agent 2) fails permanently, the system executes Agent 4 with a warning: "Intel database query offline. Verdict generated with reduced confidence parameters."
Conflicting Agent Reports (Registry matches, but GDELT reports high active local counterfeits)	Agent 4 detects positive registry match (<i>Match</i> =) but high risk warnings from Agent 2.	Agent 4 overrides positive registry flags. System marks final verdict as SUSPICIOUS / HIGH RISK with the explicit note: "Packaging details correspond to registered sequences, but active counterfeit operations are utilizing identical labeling formats. Extreme caution recommended."

8. Pedagogical Workshop Guide & Curriculum Outline

VerifyMed is designed to be built over a standard **2-Day AI Engineering Boot Camp**.

WORKSHOP ROADMAP		
DAY 1: Core Engineering	DAY 2: Orchestration	POST-CLASS: Expansion
* OCR & Ingestion	* Async execution	* Porting to local mobile
* JSON Schema design	* Synthesis Agent	* Live SQLite cache
* Basic Prompt Tuning	* Real-time Streaming UI	* Integration with NAFDAC

Day 1: Visual Ingestion & Agent Prompt Engineering

- **Module 1:** Setting up Multimodal Llama models on Groq. Parsing packaging image uploads.
- **Module 2:** Writing structured JSON system prompts. Getting students to intentionally

break and fix model outputs using JSON Mode constraints.

- **Module 3:** Connecting models to tools. Setting up a local database of registered drugs and creating search functions for Agent 1 & Agent 2.

Day 2: Parallel Orchestration, Synthesis, and Web Deploy

- **Module 4:** Implementing async concurrency patterns (using asyncio in Python or Promise.all in JavaScript) to parallelize Agents 1, 2, and 3.
- **Module 5:** Designing Agent 4's consensus engine to resolve logic conflicts and draft safe probabilistic summaries.
- **Module 6:** Formatting the modern Tailwind web dashboard, configuring live telemetry reporting, and launching the application publicly.